The Projection Law: Model-Ensemble Errors Point at Their Binding Constraint

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Abstract

Purpose: Multi-model agreement is routinely used as confidence in materials simulation and beyond, yet models in a family share construction and training lineage. This work formalizes and tests the projection law: a model family acts as a projection operator, every well-fitted member shares one residual — a single direction in observable space — and that direction fingerprints whatever constraint binds the family. **Methods:** The theoretical core (normal-cone consensus theorem; participation-ratio gauge $\mathrm{PR}(d, \rho) = (\rho + d)^2/((\rho + 1)^2 + (d - 1))$ with explicit ribbon-collapse rate (ribbon = low-dimensional ensemble error-vector geometry, not parameter-space sloppy-model ribbons); ribbon/consensus decoupling; affine decomposition of closed affine reachable sets; local normal-cone theorem for C^1 smooth non-convex immersions; and Hoeffding entrywise concentration of the empirical second-moment matrix) is machine-checked in Lean 4 with zero unproven obligations. Two factorial experiments were pre-registered with refutation conditions committed before execution: a 4×2 architecture-by-functional crossing of MatPES-trained foundation interatomic potentials evaluated on elastic constants of 15 cubic metals, and an analysis of 12 DFT implementations on 384 equation-of-state benchmarks from the ACWF verification effort. A two-tier replication kit verifies every statistic from committed raw data (NumPy only, seconds) and re-derives the raw data from public checkpoints (deterministic within 1 GPa; the Gate-0 harness regression is bit-exact). **Results:** Foundation-model errors cluster by training functional, not architecture (S_{func}=+0.317 vs. S_{arch}=−0.093, exact permutation p=0.029, 2 of 70 labelings at the resolution floor); the registered effect-size component failed, while the rotation prediction was confirmed on Au and Pt under r^{2}SCAN but not Ag. DFT implementation errors cluster by pseudopotential table, not simulation code (S_{table}=+0.526 vs. S_{code}=+0.265, p=0.017); four independent plane-wave codes sharing one table align at 0.70–0.87. Neither registered refutation condition was triggered, but four of seven registered predictions failed and are reported as failures. Combined with the 559-potential classical baseline (within-family reference–prediction correlation 0.95; error dimensionality invariant over forty years), error anisotropy is conserved across paradigm replacements while its direction rotates to the next constraint upstream. A correction direction fitted with the corrected model held out removes a median 69% of foundation-model squared elastic error out-of-sample (IQR 35–92%; per-element medians up to 97%). **Conclusion:** Agreement among models sharing a constraint measures the constraint, not the truth. Error-vector geometry converts benchmarking from scoring into diagnosis, supplies a zero-cost trust rule for ensemble uncertainty, and enables family-level calibration without retraining.

Keywords: model ensembles; error geometry; uncertainty quantification; interatomic potentials; foundation models; density functional theory; verification and validation; benchmarking; formal methods

1 Introduction

Every field that builds models in families uses inter-model agreement as evidence. The practice has long been questioned qualitatively: climate ensembles share construction and history, so their consensus may reflect common structure rather than truth [1, 2], and error correlations between members quantify that dependence [3–5]. In machine learning, independently trained networks agree far beyond what their accuracy predicts [6, 7], ensemble members cluster in prediction space [8], and disagreement tracks the shared bias of the training procedure rather than correctness [9]. What these literatures establish qualitatively, this paper makes geometric, formal, and—in the specific sense of identifying *which* constraint binds—operational.

The central object is the *error vector*: for model i in family F evaluated on observables with reference values, $e_i \in \mathbb{R}^d$ is the vector of (relative) errors. The law has three parts:

(i) Projection. A model family is a projection operator. Fitting, however performed and by whomever, drives predictions toward the point of the family's reachable set nearest the truth; the residual is a property of the *(family, target)* pair, not of any individual model or modeler. All well-fitted members therefore share one residual direction (Theorems 1–2).

(ii) Gauge. The ensemble's error second moment is a shared bias plus fitting noise; its participation ratio is a closed-form gauge of the systematic fraction (Theorem 3), so low error dimensionality is not a curiosity but a necessity for any converged family, at an explicit rate.

(iii) Conservation and rotation. When a modeling paradigm is replaced — when the binding constraint is removed — the error anisotropy does not dissolve into isotropic scatter. It is conserved, and its direction rotates to the next constraint upstream in the epistemic stack (Fig. 1a). This is the part of the law for which we find no antecedent in the climate, forecasting, machine learning, or materials literatures, and it is the part we test hardest (Fig. 1b).

We instantiate the law at three layers of a single epistemic stack — classical interatomic potentials, foundation machine-learned interatomic potentials (MLIPs), and the density-functional-theory (DFT) implementations that generate their training data — because this stack offers what few domains can: large open model ensembles, factorial structure (the same architecture trained on two functionals; the same simulation code run with two pseudopotential families), and references at every layer. The empirical discovery paper for the first layer is reported separately [10]; here that layer serves as the observational base of a cross-layer test.

Three methodological commitments distinguish this work. The theory core is *machine-checked*: every theorem below is verified in Lean 4 with zero unproven obligations, and a written contract maps each theorem to the statistic that instantiates it. The two new experiments are *pre-registered*, with thresholds and explicit refutation conditions committed to version control before any model was executed; we report all threshold misses as failures. And the entire evidence chain is packaged for *tiered replication*: the analysis layer verifies from committed raw data with no machine-learning dependencies in seconds, and the physics layer re-derives the raw data from public checkpoints deterministically (within 1 GPa tolerance; the harness regression itself is bit-exact).

2 Theoretical framework

The mathematics of best approximation is classical [11, 12]; we claim only its epistemic application to model ensembles and the machine-checked instantiation chain. Throughout, E is a real inner-product space (observable space), $s \subseteq E$ a model family's reachable set, $T \in E$ the truth.

Theorem 1 (Normal-cone criterion). *Let s be convex and p a best approximation of T in s (i.e. $p \in s$ and $\|T - p\| \leq \|T - q\|$ for all $q \in s$). Then the residual $T - p$ lies in the normal cone of s at*

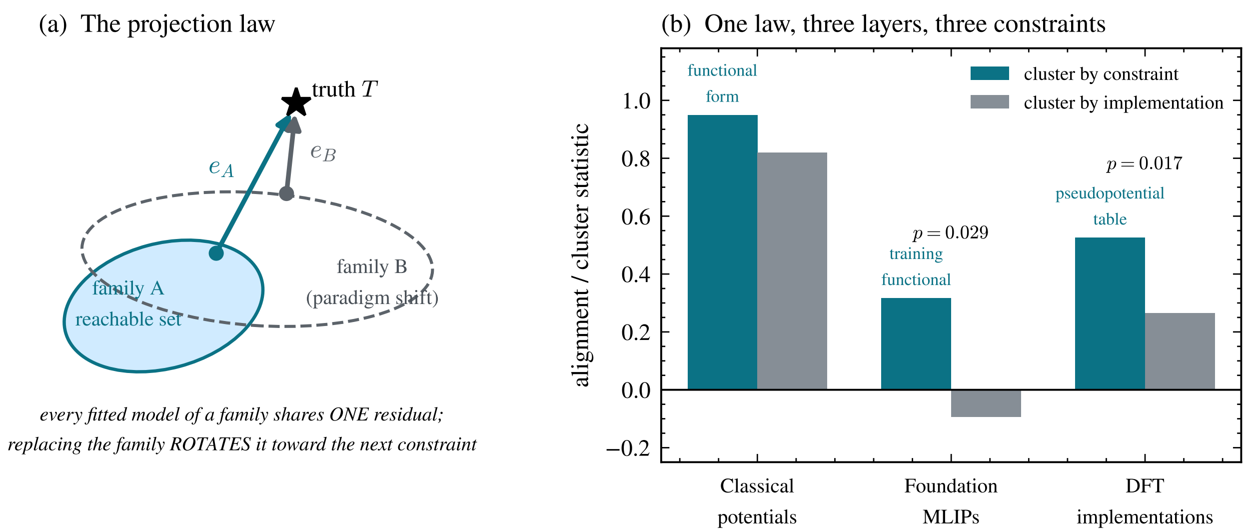

Figure 1: **The projection law and its three-layer test.** (a) A model family's reachable set fixes the residual of every well-fitted member (Theorems 1–2); replacing the family (paradigm shift) rotates the shared residual rather than removing it. (b) Cluster-by-constraint versus cluster-by-implementation statistics at three layers of one epistemic stack. Layer 1 shows within-family vs. pooled reference–prediction correlation (observational); layers 2–3 show the pre-registered cluster statistics with exact permutation p-values; the registered refutation condition (implementation > constraint) was not triggered in either experiment.

p: $\langle T - p, q - p\rangle \leq 0$ for every $q \in s$.

Theorem 2 (Consensus)**.** *Best approximations onto a convex family are unique; consequently any two best approximations of the same target share an* identical *residual. The residual — hence any agreement among independently fitted members — is determined by the (family, target) pair alone, and vanishes exactly when the truth lies in the family.*

Agreement among models sharing a constraint is therefore a measurement of the constraint, not evidence of truth: the formal content of the qualitative warnings of [2] and [1], and a strengthening of error-correlation dependence measures [3] in that, combined with the factorial designs of §4–5, it identifies *which* constraint binds.

Theorem 3 (Gauge)**.** *Model errors $e = b + \xi$ with shared bias b and isotropic noise of scale σ in d dimensions have error second moment with one eigenvalue $\|b\|^2 + \sigma^2$ (bias axis) and $d-1$ eigenvalues σ^2, and participation ratio*

$$\mathrm{PR}(d,\rho) \;=\; \frac{(\rho+d)^2}{(\rho+1)^2 + (d-1)}, \qquad \rho = \|b\|^2/\sigma^2,$$

with $\mathrm{PR}(d,0) = d$, $1 \leq \mathrm{PR} \leq d$, strict decrease in ρ, and the explicit collapse bound $\mathrm{PR} - 1 \leq 3(d-1)/\rho$ for $\rho \geq d$.

The algebra is a one-line corollary of spiked-covariance theory [13] and the participation ratio is a standard effective-dimension metric [14]; the value here is the consilience it enables (§3) and the machine-checked monotonicity that licenses inverting measured PR into a systematic fraction $\alpha = \rho/(\rho + 1)$.

Theorem 4 (Ribbon/consensus decoupling)**.** *The participation ratio of a shared-axis ensemble $\{c_i u\}$ is* 1 *for every sign pattern of the coefficients c_i, while mean pairwise alignment distinguishes them (e.g.* 1 *vs.* $-1/3$ *for three members). PR detects the* axis*; alignment detects* sign coherence*; they are distinct order parameters. ("Ribbon" here refers throughout to the error-vector geometry of a model* ensemble *in observable space, not to the parameter-space hyper-ribbons of single models [15, 16]; §3 details the distinction.)*

Theorem 5 (Affine decomposition)**.** *Let $K = a + L$ be a closed affine reachable set and p^* the best approximation of T in K. For any fitted $p \in K$, the residual decomposes as*

$$T - p \;=\; b + \xi(p), \qquad b = T - p^* \in L^{\perp}, \quad \xi(p) = p^* - p \in L,$$

with $b \perp \xi(p)$.

Theorem 5 turns the bias-plus-within-family *split* of Theorem 3 from a modeling assumption into a derivable consequence for affine families. The isotropic-noise gauge itself remains an empirical regularity: the theorem gives the shared bias direction and the orthogonal within-family subspace, but the additional assumptions of isotropic scatter and a common bias for every ensemble member are justified by the data in §3–5, not derived here.

Theorem 6 (Local normal cone, smooth non-convex families)**.** *Let $f : \mathbb{R}^k \to E$ be a C^1 immersion and x^* a local minimizer of $\|T - f(x)\|$. Then the residual $T - f(x^*)$ is orthogonal to the tangent space* range $Df(x^*)$.

Theorem 6 licenses testing tangent-space orthogonality for neural-network reachable sets: the conclusion is pointwise (one normal space per local minimizer), not the global consensus theorem. The empirical clustering in §4 is interpreted as nearby local minimizers landing in a common normal direction; it does not follow from Theorem 6 that every local minimizer shares the same residual.

Theorem 7 (Finite-sample concentration)**.** *Let $X_1, \ldots, X_n$ be i.i.d. bounded random vectors in $\mathbb{R}^d$ ($\|X_k\|_\infty \le B$) and let M and $\widehat{M}_n$ be the population and empirical entrywise second-moment matrices. For every entry (i,j) and $\varepsilon > 0$,*

$$\mathbb{P}\big(|\widehat{M}_{n,ij} - M_{ij}| \ge \varepsilon\big) \;\le\; 2\exp\!\big(-n\varepsilon^2/(2B^4)\big).$$

Moreover, the participation ratio $\mathrm{PR}(M) = (\operatorname{tr} M)^2/\operatorname{tr}(M^2)$ *is continuous wherever the denominator is non-zero.*

Theorem 7 bounds each entry of the empirical second-moment matrix; continuity of the participation ratio ensures that small entrywise perturbations cannot produce a discontinuous jump. It does not, by itself, give a closed-form sample-complexity bound for $|\widehat{\mathrm{PR}} - \mathrm{PR}|$ because the denominator $\operatorname{tr}(M^2)$ can be arbitrarily small. In practice, PR uncertainty is quantified by the bootstrap intervals reported in §3.

All seven theorems are verified in Lean 4 against a pinned Mathlib as part of the program's larger formal corpus — 225 theorem and lemma declarations across the current build, checked by `lake build` with zero `sorry` and zero new axioms — including a conditional hyper-ribbon bound (spectral-decay hypotheses $\Rightarrow$ PR < 2), a context-correction theorem bundle whose "correction closes the residual" statement is the formal counterpart of the out-of-sample calibration result below, and a formal audit module that previously caught an overstatement in the program's own empirical work. The artifact and the theorem-to-statistic contract ship with the replication kit (§10). Figure 2 visualizes the gauge with its classical inversion and the decoupling of the two order parameters in live data.

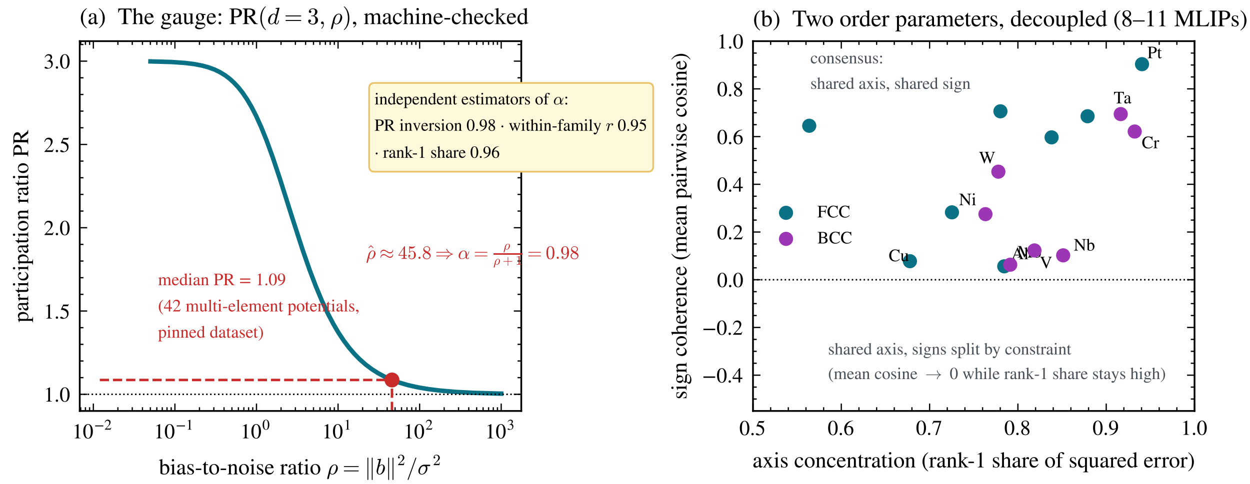

Figure 2: **The gauge and the two order parameters.** (a) PR$(3, \rho)$ (Theorem 3); inverting the classical ensemble's median PR = 1.09 gives systematic fraction $\alpha = 0.98$, in agreement with two independent estimators (§3). (b) Per-element foundation-MLIP ensembles (n = 8–11 models): axis concentration stays high for every element while sign coherence spans the full range — the decoupling of Theorem 4. Elements whose errors split in sign by training functional (V, Nb, W) sit low-right; consensus elements (Pt, Ta, Au) sit top-right.

3 Layer 1: classical interatomic potentials

The observational base [10]: elastic-constant (C_{11}, C_{12}, C_{44}) errors of 559 classical potentials across 15 cubic metals. Three facts matter for the law. First, error dimensionality is low and *stationary*: participation ratios of the 42 multi-element potentials occupy $[1.00, 2.29]$ out of 3 with median 1.09 (dataset pinned in the replication kit), with no trend across forty years of potential development — accuracy improved; geometry did not. You cannot fit your way off a constraint surface. Second, within a functional-form family the errors are nearly identical (within-family reference–prediction correlation $r = 0.95$ across families vs. 0.82 pooled). Third, the gauge is consistent: the median PR of 1.09 inverts (Theorem 3, d=3) to a systematic fraction $\alpha \approx 0.98$, in agreement with the within-family correlation (0.95) and the rank-one share (0.96). We emphasize that these three diagnostics are functionals of closely related second-moment structure on overlapping data — under the bias-plus-noise model they are algebraically coupled — so their agreement is internal consistency, not independent corroboration. The binding constraint at this layer is the functional form itself; the oldest example is exact: central-force pair potentials satisfy the Cauchy relation $C_{12} = C_{44}$ identically [17], so for any material violating it, every pair potential's error necessarily contains the same forced component — the constraint dictates the direction.

We emphasize the distinction from parameter-space sloppiness: hyper-ribbon geometry was introduced for the parameter manifolds of *single* models [15, 16, 18, 19]. The object here is different — error vectors of an ensemble of *distinct* models in observable space — though the two pictures are related through the projection of parameter manifolds into data space.

4 Layer 2: foundation MLIPs — functional vs. architecture

Foundation MLIPs replace the functional-form constraint with expressive neural networks trained on DFT corpora. The law predicts the anisotropy survives and rotates: the binding constraint becomes

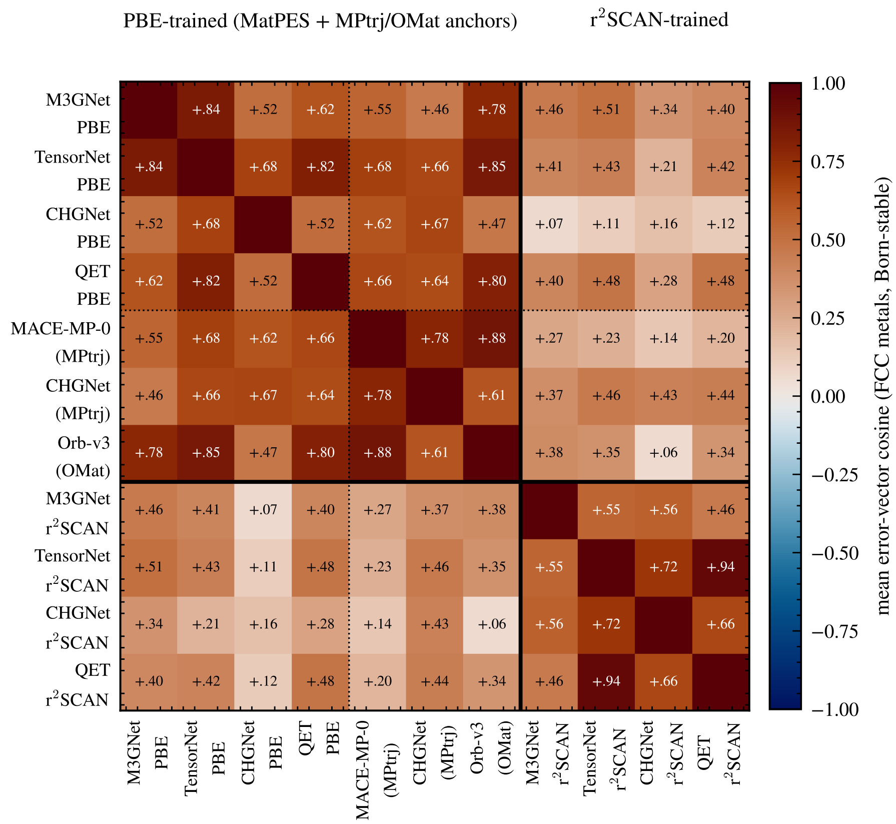

Figure 3: **Foundation-MLIP error geometry organizes by training functional, not architecture.** Mean pairwise error-vector cosine over Born-stable FCC elements for eleven models: four architectures trained on MatPES-PBE, three PBE-lineage anchors (MPtrj/OMat training, dotted separator = dataset axis with functional held fixed), and the same four architectures trained on MatPES-r^{2}SCAN (solid separator). The PBE block coheres across architectures *and* datasets; the cross-functional blocks fade; within-r^{2}SCAN pairs re-cohere (TensorNet$\times$QET $+0.94$).

the training reference theory. Qualitative inheritance of Perdew–Burke–Ernzerhof (PBE) bias — systematic softening shared by universal MLIPs — has been reported [20, 21]; the contributions here are the directional quantification and the factorial attribution.

Observation. Three PBE-lineage models of unrelated architecture (MACE-MP-0 [22], CHGNet [23], Orb-v3 [24]) commit *the same* elastic-constant errors on FCC metals against experimental references: all-negative, C_{44}-dominant (Au: $-32\%, -27\%, -79\%$), with cross-architecture cosines 0.95–0.99 for Au, Pt, Ta, Ag. An internal control excludes harness artifacts: Ni, through the identical pipeline, has near-zero error. Because every model is compared to one shared reference per element, common-mode reference error inflates *absolute* cosines; the factorial contrasts below difference it out, so the clustering statistics are immune to this effect even where the raw alignment values are not. Across 8–11 models per element the rank-one share of squared error is 0.56–0.94 — and where models "disagree" (V, Nb, W), they disagree by *sign along the same axis* (V: cosine -0.88 between its Born-stable pair), exactly the structure Theorem 4 distinguishes.

Pre-registered factorial test. The MatPES release [25] provides one curated training distribution in two functionals (PBE, r^{2}SCAN) with four architectures trained on each (M3GNet, TensorNet, CHGNet, QET) — a 4×2 crossing of constraint against implementation. We registered thresholds and a refutation condition ("errors cluster by architecture") before executing any model, ran all eight cells through one Born-screened strain-energy harness, and obtained: clustering by functional $S_{\text{func}} = +0.317$ (exact permutation $p = 0.029$, i.e. 2 of all 70 labelings; the lattice's resolution floor is 1/70) versus clustering by architecture $S_{\text{arch}} = -0.093$; the registered rotation prediction (2-of-3 criterion) confirmed on Au (C_{44} error $-0.33 \rightarrow -0.07$) and Pt ($-0.46 \rightarrow -0.31$) under r^{2}SCAN, while Ag — whose PBE error was already near zero — did not move (-0.02); and the dataset-control confirmed (MatPES-PBE models align with MPtrj/OMat-PBE anchors, median cosine 0.66). Two of the four registered predictions failed: the within-functional consensus median (0.654 vs. 0.70) and, notably, the *effect-size component of the clustering prediction itself* (0.085 vs. the registered 0.30) — the clustering direction is significant but $\sim 3.5\times$ weaker than predicted, and we report that conjunctive test as a failure. Two referee-driven robustness checks hold: without Born screening the ordering persists ($S_{\text{func}} = +0.307$ vs. $S_{\text{arch}} = -0.081$ on FCC; $+0.179$ vs. $+0.018$ over all 15 elements unscreened), so the screen did not manufacture the result; and the practical corollary survives out-of-sample — fitting the shared direction with the corrected model held out removes a median 69% of squared error across 154 leave-one-model-out trials (IQR 35–92%; per-element medians up to 97%). The data also revealed *nested* constraints — Cu, Ni, Al errors reorganize by functional (between-functional cosine down to -0.49, the predicted crossover), while the $5d$ noble metals retain a shared direction across both functionals, bound by correlation physics deeper than the PBE/r^{2}SCAN distinction. The refutation condition was not triggered.

5 Layer 3: DFT implementations — pseudopotential vs. code

One layer further up, the functional itself is held fixed. The ACWF verification effort [26, 27] provides equation-of-state parameters (V_0, B_0, B_1) for 384 unary crystals from twelve pseudopotential-based method configurations and two all-electron codes, all PBE. (The DFT calculations are public and pre-existing; our analysis was registered after downloading the data but before computing any error-vector statistic.) Against the all-electron average, with the FLEUR–WIEN2k split as the per-system noise floor, the law predicts errors organize by *pseudopotential table* (the approximation actually shared), not by *simulation code* (the implementation). The dataset's factorial structure makes the test sharp: ABINIT appears with three pseudopotential families, Quantum ESPRESSO and CASTEP with two each, while the PseudoDojo-0.4 table is implemented by four independent plane-wave codes.

Pre-registered outcome: same-table–different-code alignment $S_{\text{table}} = +0.526$ versus same-code–different-family $S_{\text{code}} = +0.265$; separation $+0.261$, permutation $p = 0.017$; refutation condition (clustering by code) not triggered. At pair level, four independent codes sharing PseudoDojo-0.4 align at 0.70–0.87, while ABINIT-with-JTH versus ABINIT-with-PseudoDojo — the same code — drops to 0.21. Two auxiliary predictions failed as registered, both informatively: the within-table consensus median (0.459 vs. 0.50) was dragged down solely by SIESTA, the one localized-orbital-basis code in the group, whose binding constraint is evidently the basis set rather than the shared pseudopotential — the nested-constraint phenomenon again; and the registered regime prediction had the wrong sign because between-family divergence grows on difficult elements, which the pooled statistic conflated. A secondary observation sharpens the law: PAW codes with *different* PAW tables align at only $+0.20$ — the constraint is the specific frozen-core construction, not the formalism label. The result is robust to the referee-relevant analysis choices: dropping the fit-noise-prone B_1 observable *increases* the separation ($+0.261 \rightarrow +0.281$), and whitening each observable by its own all-electron split increases it further ($\rightarrow +0.359$). Scalar reproducibility metrics [26, 27] cannot see any of this structure; it lives in the directions.

Table 1: The projection law across one epistemic stack. Each row: an ensemble of models, the constraint the law identifies as binding, the clustering evidence (constraint vs. implementation), and the registered refutation condition's status.

Layer	Ensemble	Binding constraint	Evidence	Kill condition
Classical IPs	559 potentials	functional form	within-family r=0.95; PR invariant 40 yr	(observational)
Foundation MLIPs	4×2 MatPES + anchors	training functional	S_{func}=+0.317 vs −0.093, exact perm. p=0.029 (2/70, floor)	not triggered
DFT codes	12 ACWF methods	pseudopotential table	S_{table}=+0.526 vs +0.265, p=0.017	not triggered

6 The conservation–rotation law

Table 1 assembles the stack. Three layers, three different binding constraints, one geometric law; the two new layers tested under pre-registration with explicit kill conditions, neither triggered. Between layers, the direction *rotates*: no transfer of classical cross-family alignment to MLIP alignment is detectable (Spearman $\rho = 0.26$, $p = 0.34$ across elements — a low-powered test, so this is absence of evidence for transfer rather than evidence of independence; the rotation claim rests on the factorial clustering, not on this null), because the constraint moved from functional form to training functional; and the MLIP-layer bias direction is precisely the reference-theory error that layer 3 isolates. Within layers, constraints *nest*: when the registered constraint is removed (r^2SCAN for PBE; plane-wave pseudopotential sharing for SIESTA), residual alignment reveals the next constraint in line (5d correlation physics; basis set). Anisotropy is conserved throughout, in the operational sense that the participation ratio stays far below the ambient dimension at every layer — and the quantitative agreement across layers is striking: median PR = 1.09 (classical, 42 multi-element potentials), 1.13–2.09 per element (foundation MLIPs, n = 8–11 models), and median 1.10 (range $[1.00, 2.11]$, rank-one share median 0.95) across 134 above-floor ACWF systems with ≥ 6 methods. At no layer, in no ensemble, under no intervention did errors approach isotropy, as Theorem 3 requires of any family whose systematic error dominates its fitting scatter.

The closest antecedents are the persistence of shared biases across climate model generations [5, 28], the convergence of training trajectories onto shared low-dimensional manifolds [29], and representation convergence across architectures [30]; none states conservation-plus-rotation as a law, and none possesses the factorial constraint-vs-implementation evidence presented here.

7 Related work

Ensemble dependence across fields. The climate community has the deepest tradition here. [1] and [2] argued qualitatively that multi-model agreement may reflect shared construction; [3] made dependence operational through pairwise error correlation under the replicate-Earth paradigm, with the elegant null that truly independent models would correlate at 0.5 through the shared observations; [31] review the resulting weighting and sub-selection machinery, [32] implement performance-plus-independence weighting, [33] traces similarity to literal component replication across modeling centers, and [34, 35] place inter-model error in low-dimensional EOF coordinates. Our normal-cone theorem supplies the mechanism beneath these practices — dependence is not merely shared code but shared *reachable set* — and the factorial designs add what error correlation alone cannot: identification of *which* shared element binds. The persistence of the double-ITCZ bias across three CMIP generations [28] reads, in our terms, as a conservation observation awaiting its rotation experiment.

Machine learning. That independently trained networks agree beyond chance is quantified by error consistency [7] and model similarity [6]; ensemble members cluster in prediction space [8] despite weight-space diversity, which is why deep-ensemble uncertainty [36] inherits the shared component as unaccounted bias — the ambiguity decomposition of [37] already showed ensemble spread bounds only the variance term. Disagreement tracks the training procedure's shared bias [9], and accuracy- and agreement-on-the-line [38, 39] exhibit one-dimensional collective structure under distribution shift. Most complementary is underspecification [40]: it studies how equally-fitted models *differ* along directions the family and data leave unconstrained — our noise term ξ — whereas the projection law studies what they *share* — the bias b. Theorem 4 is precisely the statement that these are separate order parameters, measurable separately. Representation convergence across architectures [30] and shared training manifolds [29] are the trajectory- and representation-space cousins of our observable-space result.

Materials simulation. [41] pioneered ensembles of interatomic potentials for error estimation — spread as uncertainty; the law identifies exactly the component spread misses (the shared residual) and supplies the diagnostic for when it dominates (alignment). Shared softening of universal MLIPs is documented [20, 21], and [42] report that GGA→r^{2}SCAN transfer is hampered by poor inter-functional correlation — our factorial result gives that observation a geometric form: light-metal elastic errors genuinely cross over between functionals while noble-metal errors share a deeper constraint. DFT reproducibility is measured by scalar and curve metrics [26, 27]; the directional structure of §5 is invisible to those metrics by construction. Parameter-space sloppiness of single models [15, 16, 18, 19] is the intellectual ancestor of this work; we re-emphasize that the object here — error vectors of an ensemble of distinct models in observable space — is different from, though related to, the parameter manifold of any one of them. The participation ratio is a standard effective dimension [14] and the gauge algebra is spiked-covariance theory [13]; the best-approximation mathematics is classical [11, 12].

What is new here is the combination: the error-vector geometry of model ensembles as the measured object; factorial constraint-vs-implementation designs with pre-registered refutation conditions at two layers of one stack; the conservation–rotation law, which none of the above states; and a machine-checked theory chain bound to the measurements by a written contract.

8 Discussion: implications for benchmarking, VVUQ, and calibration

Calibration without retraining. Because the shared error is one direction, one correction vector per (element, observable) repairs every model in the family at once — and the claim survives the obvious circularity objection: fitting the direction with the corrected model *held out* removes a median 69% of squared elastic error out-of-sample (154 leave-one-model-out trials, IQR 35–92%; per-element medians reach 97% where the constraint binds hardest). The correction is family-level — it transfers to models not yet trained, so long as they share the constraint.

A trust rule for ensemble UQ. Cross-model alignment is a zero-cost diagnostic: high alignment ⇒ the ensemble is constraint-bound, calibratable, and its spread *understates* uncertainty; low alignment at the noise floor ⇒ genuine convergence (the Ni control); low alignment far above the floor ⇒ the constraint is heterogeneous (magnetic BCC metals; difficult elements across pseudopotential tables) and no shared correction exists.

Benchmark design. Leaderboards that pool across families average away the very direction that identifies what to fix. Reporting error *directions* stratified by candidate constraints — the factorial pattern used here — converts benchmarking from scoring into diagnosis. This is directly actionable for materials data infrastructure: benchmark records should carry the error vector, not only its norm.

Reading consensus. Where ground truth is delayed (materials discovery campaigns, extrapolative simulation), inter-model agreement should be priced as a measurement of shared constraint; the factorial designs show how to estimate *which* constraint, today, from models alone.

9 Limitations

The formal chain covers convex reachable sets and deterministic bias-plus-noise spectra; smooth non-convex families (local normal cones) and the finite-sample step from noisy ensembles to the exact second moment are standard but unformalized. The MLIP layer uses one validated harness (energy-based, $\varepsilon_{\max} = 0.5\%$, float32, non-spin-initialized inference) on fifteen cubic elemental metals and elastic observables — the lowest-dimensional, most symmetric corner of materials space — and compares FCC metals to *experimental* references and BCC/non-FCC metals to Materials Project (DFT-PBE) references, so the measured residual mixes fitting error, exchange–correlation bias, reference-standard differences, and thermal/zero-point offsets; a per-element 0 K DFT-PBE elastic anchor (registered for round 2) is the decisive test that would separate them, and for the magnetic elements (Fe, Cr, Ni, V) the spin protocol is a live confound. The DFT layer inherits ACWF's protocol choices, and the SIESTA nested-constraint reading is post-hoc until an independent localized-basis code is tested. Statistically: the permutation lattices are small (minimum attainable $p = 1/70$), seven registered predictions across two experiments invite multiplicity concerns that round 2's single-primary-endpoint design addresses; in the present manuscript the two kill conditions were primary (cluster-by-architecture for MLIPs, cluster-by-code for DFT), while the other five predictions were auxiliary robustness checks with no formal hierarchical correction applied, and the registered kill conditions were asymmetric (refutation required a sign reversal; round 2 registers a symmetric equivalence bound). Four of the seven registered predictions failed and are reported as failures (2/4 and 1/3 passing per experiment); the nested-constraint attributions of those misses are hypotheses for the next registration round, not confirmed claims. The law itself makes its next predictions cheap to test: an r^{2}SCAN-trained model from a fourth architecture must join the r^{2}SCAN cluster; a plane-wave implementation of any new pseudopotential table must align with its table, not its code; and axis-based statistics (mandated by Theorem 4) should replace signed-cosine thresholds.

10 Reproducibility

Both experiments were pre-registered with thresholds and refutation conditions committed to version control before model execution for the 4×2 (commit `dffbe595`) and before any error-vector statistic was computed for the ACWF analysis (commit `ebf39e33`); all factorial statistics are replayable from a two-tier kit in which Tier 1 verifies the factorial-experiment statistics from committed raw data using only NumPy/SciPy in seconds (classical-layer and robustness statistics are reproduced by accompanying scripts in the same kit), and Tier 2 re-derives the raw elastic constants from public checkpoints through a frozen harness (deterministic within 1 GPa; the Gate-0 harness regression is bit-exact). The complete kit, pinned datasets, and pre-registrations are publicly served at https://storage.googleapis.com/shed-489901-replication/error-geometry/

v1-10c18ace/index.html; the citable Zenodo snapshot is archived at https://doi.org/10.5281/zenodo.20787874. The Lean 4 artifact (225 theorem and lemma declarations in the current build, including the seven theorems of this paper; zero `sorry`, zero new axioms, pinned Mathlib) and the theorem-to-statistic contract accompany the kit. The harness was validated by independent energy-vs. stress-method agreement ($\leq 2.4\%$), bit-exact regression, and strain-window stability. ACWF data are from the public archive of [26]; MatPES checkpoints from [25].

Declarations

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Data and code availability. The two-tier replication kit (raw data, analysis scripts, frozen harness), the pre-registration documents with their commit hashes, the Lean 4 formal artifact, and figure-generation scripts are available in the project repository and archived on Zenodo at https://doi.org/10.5281/zenodo.20787874. Third-party data: ACWF verification archive [26]; MatPES checkpoints [25]; OpenKIM/NIST classical-potential data as described in [10].

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References

[1] Zachary Pirtle, Ryan Meyer, and Andrew Hamilton. What does it mean when climate models agree? A case for assessing independence among general circulation models. *Environmental Science & Policy*, 13:351–361, 2010. doi: 10.1016/j.envsci.2010.04.004.

[2] Wendy S. Parker. When climate models agree: The significance of robust model predictions. *Philosophy of Science*, 78:579–600, 2011. doi: 10.1086/661566.

[3] Craig H. Bishop and Gab Abramowitz. Climate model dependence and the replicate Earth paradigm. *Climate Dynamics*, 41:885–900, 2013. doi: 10.1007/s00382-012-1610-y.

[4] James D. Annan and Julia C. Hargreaves. Understanding the CMIP3 multimodel ensemble. *Journal of Climate*, 24:4529–4538, 2011. doi: 10.1175/2011jcli3873.1.

[5] Reto Knutti, David Masson, and Andrew Gettelman. Climate model genealogy: Generation CMIP5 and how we got there. *Geophysical Research Letters*, 40:1194–1199, 2013. doi: 10.1002/grl.50256.

[6] Horia Mania, John Miller, Ludwig Schmidt, Moritz Hardt, and Benjamin Recht. Model similarity mitigates test set overuse. In *Advances in Neural Information Processing Systems 32*, 2019.

[7] Robert Geirhos, Kristof Meding, and Felix A. Wichmann. Beyond accuracy: quantifying trial-by-trial behaviour of CNNs and humans by measuring error consistency. In *Advances in Neural Information Processing Systems 33*, 2020.

[8] Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. Deep ensembles: A loss landscape perspective, 2019.

[9] Yiding Jiang, Vaishnavh Nagarajan, Christina Baek, and J. Zico Kolter. Assessing generalization of SGD via disagreement. In *International Conference on Learning Representations*, 2022.

[10] Alex Welcing. Forty years of interatomic potentials, one error geometry: hyper-ribbon universality and why pooled benchmarks mislead. Working paper, in preparation, 2026.

[11] Frank Deutsch. *Best Approximation in Inner Product Spaces*. Springer, New York, 2001.

[12] David Kinderlehrer and Guido Stampacchia. *An Introduction to Variational Inequalities and Their Applications*. Academic Press, New York, 1980.

[13] Iain M. Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29:295–327, 2001. doi: 10.1214/aos/1009210544.

[14] Peiran Gao and Surya Ganguli. On simplicity and complexity in the brave new world of large-scale neuroscience. *Current Opinion in Neurobiology*, 32:148–155, 2015. doi: 10.1016/j.conb.2015.04.003.

[15] Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Physical Review E*, 83:036701, 2011. doi: 10.1103/physreve.83.036701.

[16] Yonatan Kurniawan, Cody L. Petrie, Kinamo J. Williams, Mark K. Transtrum, Ellad B. Tadmor, Ryan S. Elliott, Daniel S. Karls, and Mingjian Wen. Bayesian, frequentist, and information geometric approaches to parametric uncertainty quantification of classical empirical interatomic potentials. *Journal of Chemical Physics*, 156:214103, 2022. doi: 10.1063/5.0084988.

[17] Max Born and Kun Huang. *Dynamical Theory of Crystal Lattices*. Oxford University Press, Oxford, 1954.

[18] Benjamin B. Machta, Ricky Chachra, Mark K. Transtrum, and James P. Sethna. Parameter space compression underlies emergent theories and predictive models. *Science*, 342:604–607, 2013. doi: 10.1126/science.1238723.

[19] Katherine N. Quinn, Michael C. Abbott, Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Information geometry for multiparameter models: new perspectives on the origin of simplicity. *Reports on Progress in Physics*, 86:035901, 2023. doi: 10.1088/1361-6633/aca6f8.

[20] Bowen Deng, Yunyeong Choi, Peichen Zhong, Janosh Riebesell, Shashwat Anand, Zhuohan Li, KyuJung Jun, Kristin A. Persson, and Gerbrand Ceder. Systematic softening in universal machine learning interatomic potentials. *npj Computational Materials*, 11:9, 2025. doi: 10.1038/s41524-024-01500-6.

[21] Haochen Yu, Matteo Giantomassi, Giuliana Materzanini, Junjie Wang, and Gian-Marco Rignanese. Systematic assessment of various universal machine-learning interatomic potentials. *Materials Genome Engineering Advances*, 2:e58, 2024. doi: 10.1002/mgea.58.

[22] Ilyes Batatia et al. A foundation model for atomistic materials chemistry, 2024.

[23] Bowen Deng, Peichen Zhong, KyuJung Jun, Janosh Riebesell, Kevin Han, Christopher J. Bartel, and Gerbrand Ceder. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 5:1031–1041, 2023. doi: 10.1038/s42256-023-00716-3.

[24] Benjamin Rhodes, Sander Vandenhaute, Vaidotas Šimkus, James Gin, Jonathan Godwin, Tim Duignan, and Mark Neumann. Orb-v3: atomistic simulation at scale, 2025.

[25] Aaron D. Kaplan et al. A foundational potential energy surface dataset for materials, 2025.

[26] Emanuele Bosoni et al. How to verify the precision of density-functional-theory implementations via reproducible and universal workflows. *Nature Reviews Physics*, 6:45–58, 2024. doi: 10.1038/s42254-023-00655-3.

[27] Kurt Lejaeghere et al. Reproducibility in density functional theory calculations of solids. *Science*, 352:aad3000, 2016. doi: 10.1126/science.aad3000.

[28] Baijun Tian and Xinyu Dong. The double-ITCZ bias in CMIP3, CMIP5, and CMIP6 models based on annual mean precipitation. *Geophysical Research Letters*, 47:e2020GL087232, 2020. doi: 10.1029/2020gl087232.

[29] Jialin Mao, Itay Griniasty, Han Kheng Teoh, Rahul Ramesh, Rubing Yang, Mark K. Transtrum, James P. Sethna, and Pratik Chaudhari. The training process of many deep networks explores the same low-dimensional manifold. *Proceedings of the National Academy of Sciences*, 121: e2310002121, 2024. doi: 10.1073/pnas.2310002121.

[30] Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[31] Gab Abramowitz, Nadja Herger, Ethan Gutmann, et al. ESD reviews: Model dependence in multi-model climate ensembles: weighting, sub-selection and out-of-sample testing. *Earth System Dynamics*, 10:91–105, 2019. doi: 10.5194/esd-10-91-2019.

[32] Reto Knutti, Jan Sedláček, Benjamin M. Sanderson, Ruth Lorenz, Erich M. Fischer, and Veronika Eyring. A climate model projection weighting scheme accounting for performance and interdependence. *Geophysical Research Letters*, 44:1909–1918, 2017. doi: 10.1002/2016gl072012.

[33] Julien Boé. Interdependency in multimodel climate projections: Component replication and result similarity. *Geophysical Research Letters*, 45:2771–2779, 2018. doi: 10.1002/2017GL076829.

[34] Benjamin M. Sanderson, Reto Knutti, and Peter Caldwell. A representative democracy to reduce interdependency in a multimodel ensemble. *Journal of Climate*, 28:5171–5194, 2015. doi: 10.1175/jcli-d-14-00362.1.

[35] Swen Brands. Common error patterns in the regional atmospheric circulation simulated by the CMIP multi-model ensemble. *Geophysical Research Letters*, 49:e2022GL101446, 2022. doi: 10.1029/2022gl101446.

[36] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems 30*, 2017.

[37] Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems 7*. MIT Press, 1995.

[38] John P. Miller, Rohan Taori, Aditi Raghunathan, Shiori Sagawa, Pang Wei Koh, Vaishaal Shankar, Percy Liang, Yair Carmon, and Ludwig Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.

[39] Christina Baek, Yiding Jiang, Aditi Raghunathan, and J. Zico Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. In *Advances in Neural Information Processing Systems 35*, 2022.

[40] Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23 (226):1–61, 2022. URL http://jmlr.org/papers/v23/20-1335.html.

[41] Søren L. Frederiksen, Karsten W. Jacobsen, Kevin S. Brown, and James P. Sethna. Bayesian ensemble approach to error estimation of interatomic potentials. *Physical Review Letters*, 93: 165501, 2004. doi: 10.1103/physrevlett.93.165501.

[42] Xu Huang, Bowen Deng, Peichen Zhong, Aaron D. Kaplan, Kristin A. Persson, and Gerbrand Ceder. Cross-functional transferability in universal machine learning interatomic potentials, 2025.

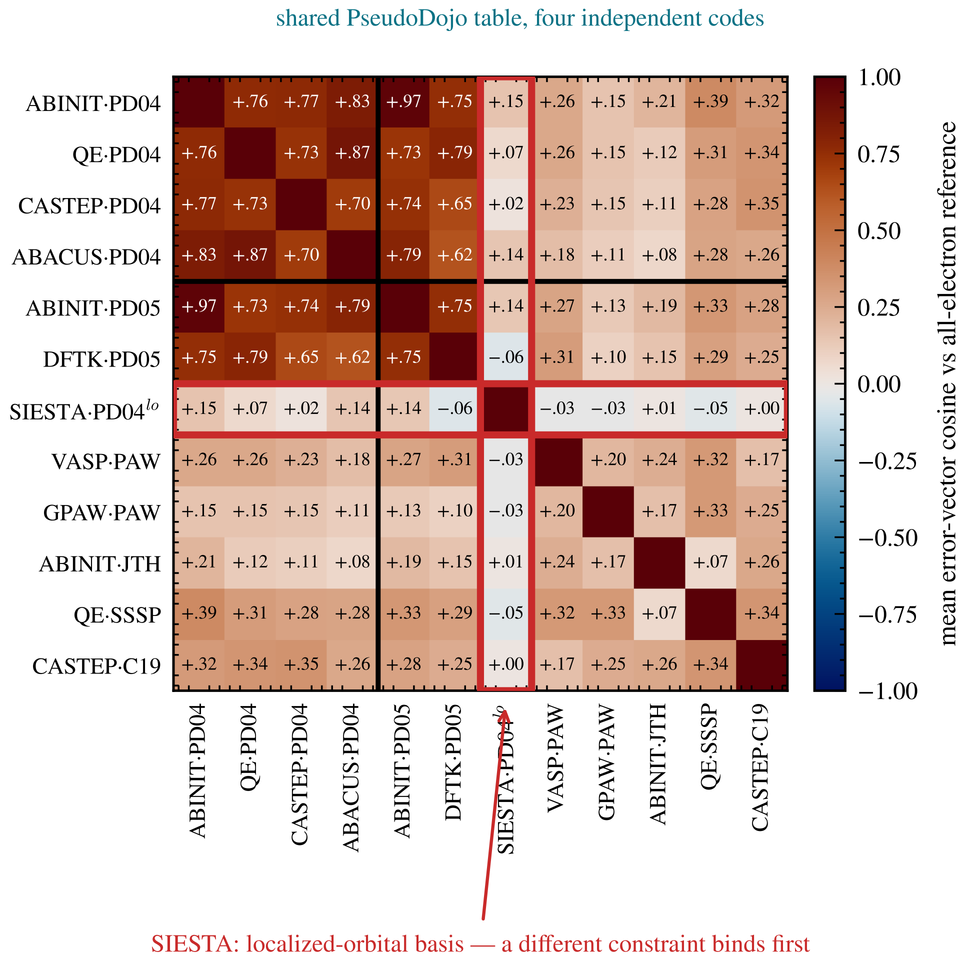

Figure 4: **DFT-implementation errors organize by pseudopotential table, not simulation code.** Mean error-vector cosine in (V_0, B_0, B_1) space against the all-electron reference, over systems above the FLEUR–WIEN2k noise floor (ACWF data, 384 unary crystals). Four independent plane-wave codes sharing the PseudoDojo table align at 0.70–0.87 (dark block); the same code under a different pseudopotential family drops to 0.19–0.35 (ABINIT-JTH vs. ABINIT-PD04 = +0.21). SIESTA (red frame), the one localized-orbital-basis method, dis-aligns from its own pseudopotential group: a different constraint binds first — the nested-constraint phenomenon.